DoRAemon's Secret Tool for an A+ in Deep Learning: Weight-Decomposed Fine-Tuning

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Introduction

Full fine-tuning large language models is expensive. **LoRA** reduces the cost by training low-rank adapter updates while freezing the base model.[1]

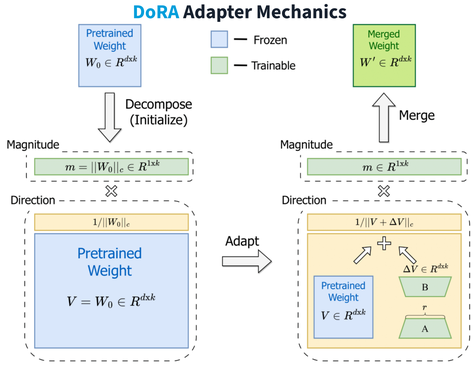
DoRA

- **DoRA** decomposes pretrained weights into **magnitude** and **direction**.[3]
- Direction receives a **LoRA**-style low-rank update; magnitude is learned separately.

Project goal

- Does **DoRA** improve commonsense accuracy over a same-scope **LoRA** baseline?
- Reproduce the LLaMA2-7B trend and test attention-only, MLP-only, and full-scope adapters.

DoRA Adapter Mechanics



Methodology

LLaMA

- Trained on 15k examples instead of 170k due to compute constraints.
- We otherwise followed the **DoRA** paper setup, reporting per-task and macro-average accuracy across all eight benchmarks.

Controlled comparison. Frozen base model, adapter-only checkpoints, 3 epochs, cutoff length 256, seed 42. Settings: **LoRA** r=32, LR=3e-4; rank-halved **DoRA** r=16, LR=2e-4.

Our Independent Experiment: Scope Ablation

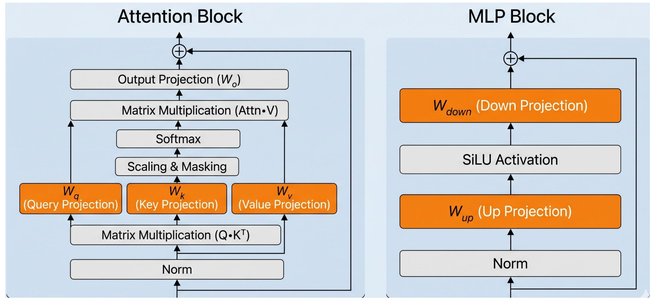


Figure 1. The ablation isolates adapter placement in attention projections, MLP projections, or both.

Scope	Target modules	Purpose
Full	`q,k,v,up,down` projections	Paper-level reproduction
Attention only	`q,k,v` projections	Test routing/context layers
MLP only	`up,down` projections	Test feed-forward layers

Verdict: DoRA Improves Accuracy and Efficiency

Accuracy	77.41%	vs	79.20%	+1.79	↑
Trainable Parameters	0.83%	vs	0.43%	-48.2%	↓

The Adapter Duel

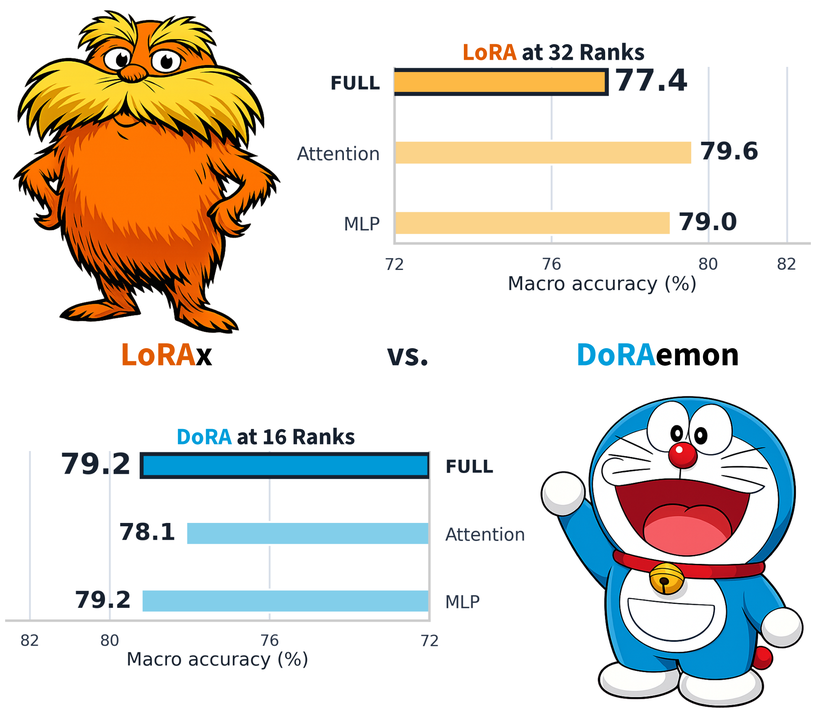


Figure 2. The full-scope bars are outlined: rank-halved **DoRA** improves macro accuracy from 77.4% to 79.2%.

Benchmark with Largest DoRA Gain: HellaSwag (+6.0 pts)

Task Format — Grounded Activity Completion

Given a short activity description, select the most plausible four-way continuation. Requires grounded physical-world commonsense.

Context: A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She pours soap on the dog and…

(A) lathers the dog up with a brush before rinsing.
(B) pours more water slowly into the bucket.
(C) scrubs the dog down and rinses it thoroughly.
(D) lets the dog run off and air dry on the lawn.

Official Paper vs Full-Scope LLaMA2-7B Reproduction

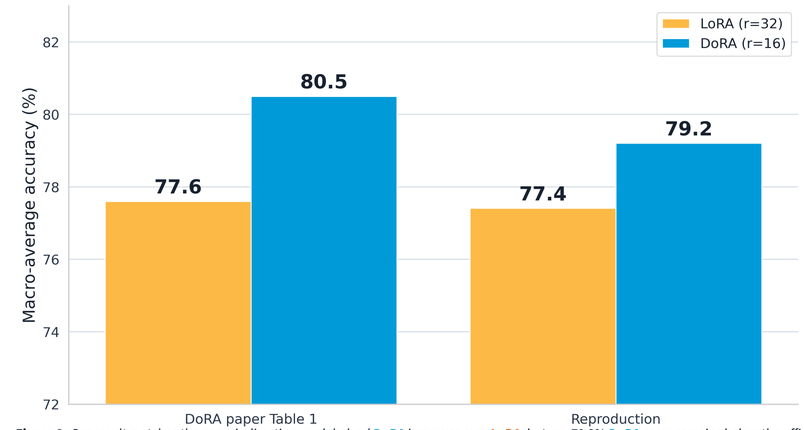


Figure 3. Our result matches the paper's direction: rank-halved **DoRA** improves over **LoRA**, but our 79.2% **DoRA** score remains below the official 80.5% reference.

LoRA vs DoRA Across Ranks

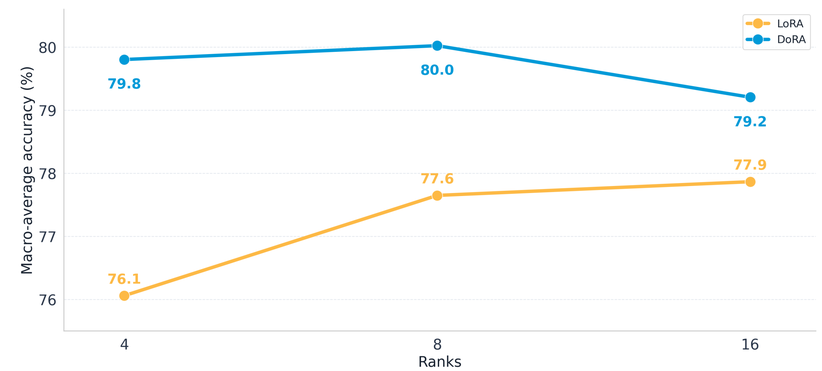


Figure 4. Macro-average accuracy across rank sizes, with the DoRA advantage shown in the lower panel.

Conclusion & Future Work

Takeaways

- Full-scope **DoRA** improved macro-average accuracy by **+1.79 points**.
- The reproduction supports the qualitative value of weight-decomposed low-rank adaptation.
- The ablation adds a caution: applying **DoRA** narrowly can remove the benefit.

Next steps

- Train on the full `commonsense_170k` dataset.
- Repeat all six conditions across multiple random seeds.
- Add parameter-count, GPU-memory, and wall-clock cost comparisons.

References

[1] Edward J. Hu et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
[2] Hugging Face. Transformers documentation. `https://huggingface.co/docs/transformers`, 2026.
[3] Shih-Yang Liu et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. Oral presentation.
[4] NVIDIA Research. NVlabs/DoRA: Official implementation. `https://github.com/NVlabs/DoRA`, 2024.